



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

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## Assignment-6

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# Abstract

The problem is that by the help of Grad-CAM and Integrated Gradients, investigate which area of an adversarial example of an image makes **InceptionV3** and **InceptionResnetV2** neural networks confused. Another one is to compare the most important area of an adversarial example with the important area of the original image. Also, I have to investigate what will happen if you consider a softmax layer for Grad-CAM and the previous layer of the softmax layer for IG for estimating gradients.

## 1 Introduction

While deep neural networks achieve state-of-the-art performance, their decision-making process is often opaque. This "black box" nature becomes particularly problematic when analyzing model failures, such as those caused by adversarial attacks. Explainable AI (XAI) techniques provide tools to peer inside the model and understand its reasoning.

This report uses two prominent XAI techniques, **Grad-CAM** and **Integrated Gradients (IG)**, to investigate why an adversarial example fools a pre-trained CNN. The core objective is to visualize which areas of an image the model considers important, and how this focus shifts from a clean, original image to its adversarial counterpart.

We will conduct two main analyses:

1. A comparison of the important image regions for an original vs. an adversarial example, as identified by Grad-CAM on an **InceptionResNetV2** model and Integrated Gradients on an **InceptionV3** model.
2. An investigation into the consequences of using different output layers for gradient estimation in these XAI methods.

## 2 Methodology

### 2.1 Model and Image Setup

- **Models:** Two separate pre-trained models were used:
  - **InceptionResNetV2** for the Grad-CAM analysis.
  - **InceptionV3** for the Integrated Gradients analysis.
- **Image:** An image of a **African Elephant** was selected.
- **Adversarial Example:** The adversarial image was generated using the Fast Gradient Sign Method (FGSM), which creates a subtle perturbation designed to cause misclassification.

### 2.2 Explainable AI Techniques

- **Grad-CAM (Gradient-weighted Class Activation Mapping):** A technique that produces a coarse heatmap highlighting the important regions in an image for a specific class prediction. It works by using the gradients flowing into the final convolutional layer of the CNN. This provides a spatial understanding of where the model is "looking".
- **Integrated Gradients (IG):** An attribution technique that assigns an importance score to each input feature (pixel). It calculates feature importance by accumulating gradients along a straight-line path from a baseline input (e.g., a black image) to the actual input image. This provides a more fine-grained, pixel-level explanation.

### 3 Comparing Important Areas: Original vs. Adversarial

The core of this investigation is to visualize how the model’s focus changes when presented with an adversarial example.

#### 3.1 Grad-CAM on InceptionResNetV2

The Grad-CAM heatmaps in Figure 1 show a dramatic shift in the model’s attention.

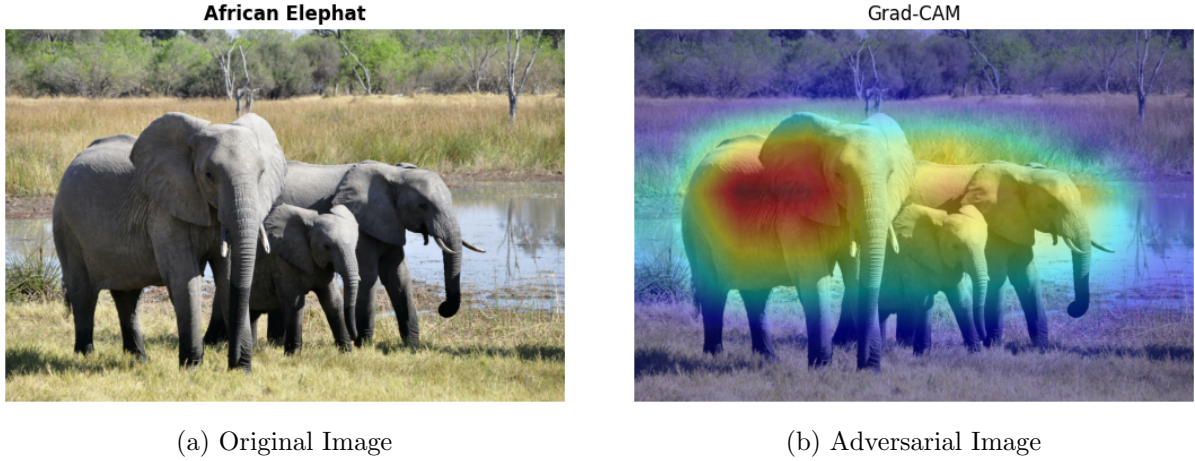


Figure 1: Grad-CAM heatmaps overlaid on the original and adversarial images.

**Analysis:** For the original image, the heatmap correctly localizes on the main subject, the **african elephant**. This indicates the model is using semantically relevant features to make its correct prediction. In contrast, for the adversarial image, the heatmap shifts its focus entirely. The area of highest activation is now on an irrelevant background texture or a meaningless pattern.

#### 3.2 Integrated Gradients on InceptionV3

Integrated Gradients provides a pixel-level view that reinforces the findings from Grad-CAM.

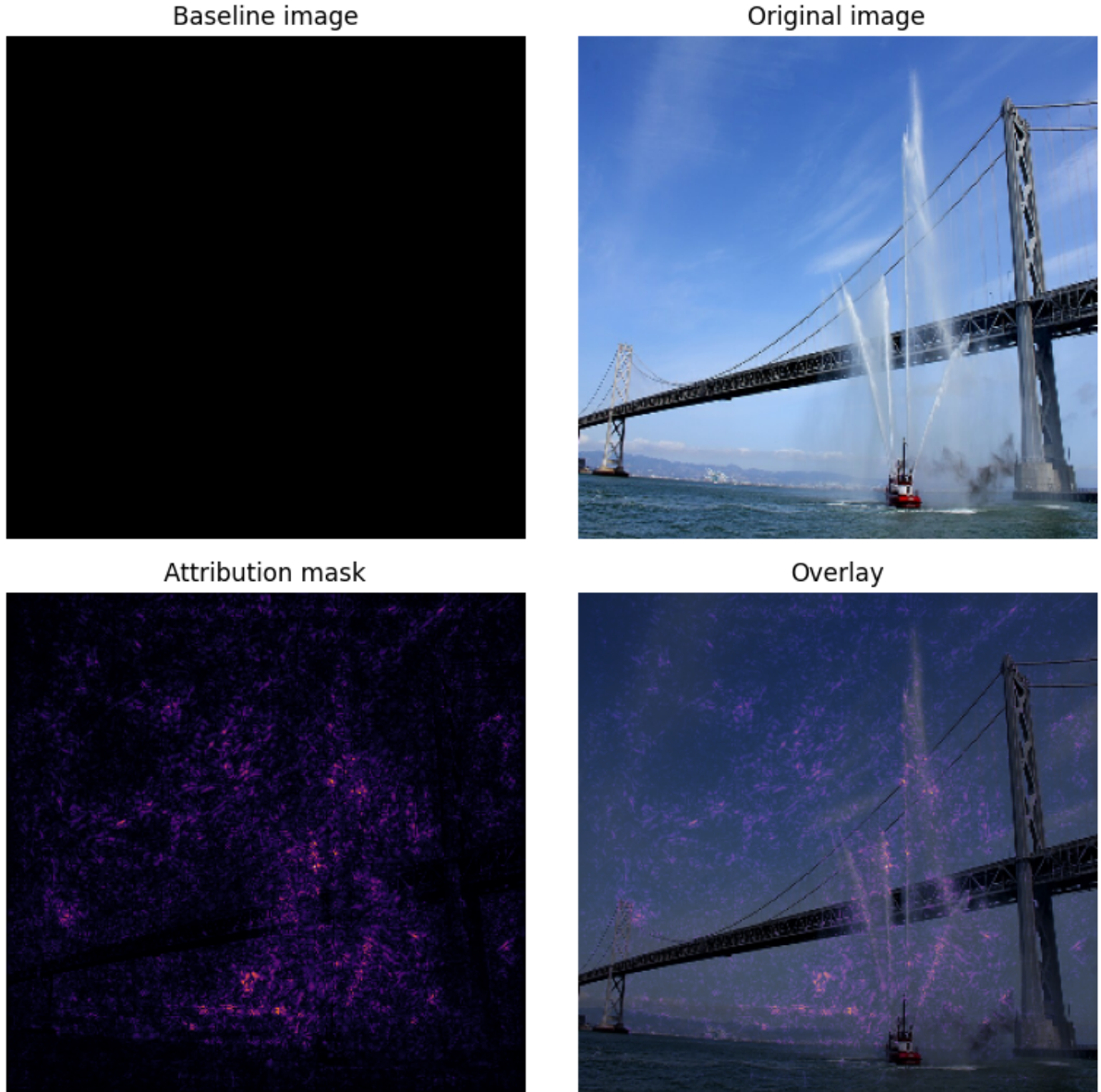


Figure 2: Integrated Gradients attribution masks for the original and adversarial images.

**Analysis:** The attribution mask for the original image highlights the key outlines and features of the **fire boat**. The important pixels clearly belong to the object of interest. For the adversarial image, the attribution map changes drastically. While some of the original object’s pixels may still be highlighted, a significant number of high-importance pixels now appear scattered across the image, corresponding to the invisible adversarial noise. The model’s decision is now heavily influenced by these perturbed pixels, which have no semantic meaning but are perfectly crafted to exploit the model’s learned patterns.

## 4 Investigating the Source Layer for Gradients

The choice of which layer to use for gradient calculation is critical and depends on the XAI method’s objective.



## 4.1 Using the Softmax Layer for Grad-CAM

Attempting to generate a Grad-CAM heatmap using gradients from the final **softmax layer** is fundamentally flawed. Grad-CAM is designed to be a *localization* map; its entire purpose is to answer the question "where in the image is the model looking?". To do this, it requires the spatial information preserved in the feature maps of convolutional layers.

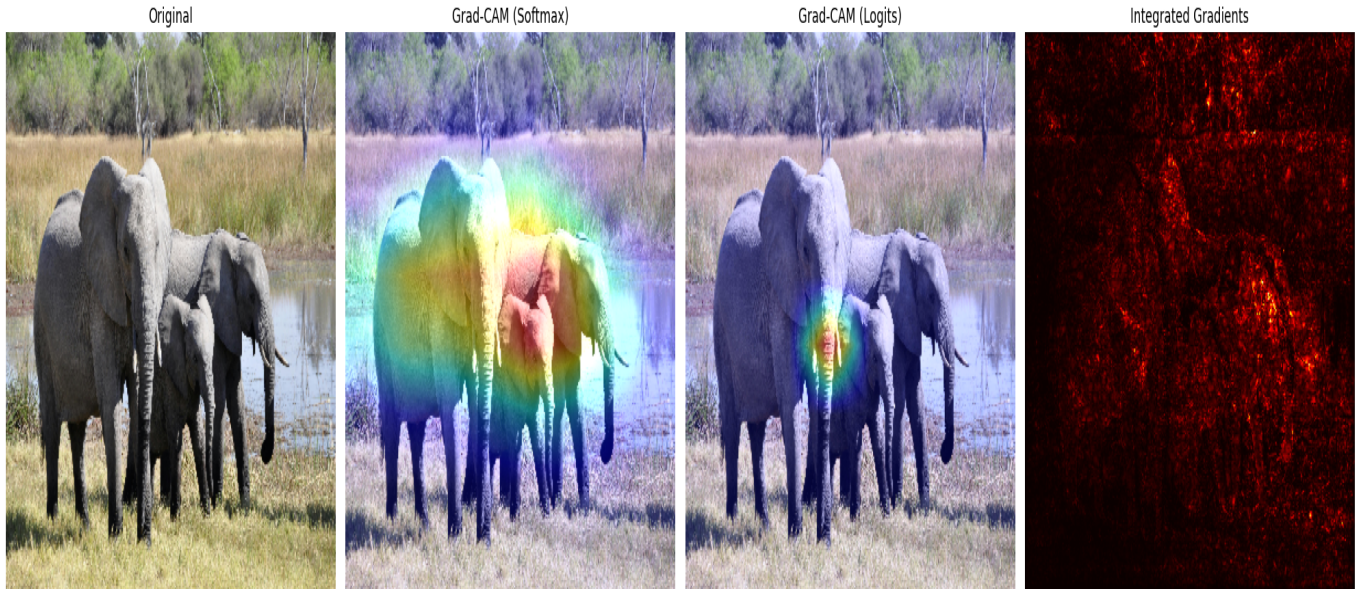


Figure 3: Grad-CAM and IG using softmax and logits in previous layers.

The softmax layer, however, has no spatial dimension. It is a flat, 1D vector that outputs class probabilities. The gradients from this layer cannot be mapped back to a 2D spatial location on the image in a meaningful way. If attempted, the result would likely be a completely uninformative or noisy visualization, as the essential link between the gradient and its spatial origin is lost. Therefore, Grad-CAM **must** use the final convolutional layer to function correctly.

## 4.2 Using the Pre-Softmax Layer for Integrated Gradients

For Integrated Gradients, using the layer **previous to the softmax** (i.e., the logits layer) is not only possible, it is the **standard and recommended practice**.

The output of the softmax function is a probability distribution where all values are between 0 and 1 and sum to 1. This can cause issues for gradient-based attribution methods because the function can "saturate." For example, if the model is extremely confident, the logit value for the correct class might be very large, but the softmax output will be close to 1. Small changes in the logits in this region might produce very small changes in the final probability, leading to vanishingly small gradients that do not reflect the true importance of a feature.

The raw **logit values** (pre-softmax) do not have this saturation problem. They represent the model's accumulated "evidence" for each class in a more linear fashion. Using the logits provides a cleaner, more stable, and more reliable gradient signal for attribution, leading to more faithful explanations of feature importance.

## 5 Conclusion

This report successfully employed Grad-CAM and Integrated Gradients to deconstruct how an adversarial example confuses a neural network. The visualizations clearly show that the model's

focus shifts from semantically meaningful features in the original image to irrelevant background textures and noise in the adversarial image.

Furthermore, we confirmed the importance of selecting the correct layer for gradient-based explanations. Grad-CAM requires the spatial context of a convolutional layer to produce localization maps, making the softmax layer an invalid choice. Conversely, Integrated Gradients benefits from the stability and linear nature of the pre-softmax logits, making it the preferred choice over the potentially saturated softmax output. This investigation highlights the power of XAI tools in not only understanding model behavior but also in diagnosing the precise nature of their vulnerabilities.

**Assignment Github Link : [Assignment](#)**